From Censorship to Detoxification: Rewriting Harmful Memes

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Problem definition

Harmful memes combine images and overlaid text to spread offensive content across both online platforms and physical spaces. Existing moderation relies mainly on **censorship** after exposure: removing, blurring, or suppressing the content.

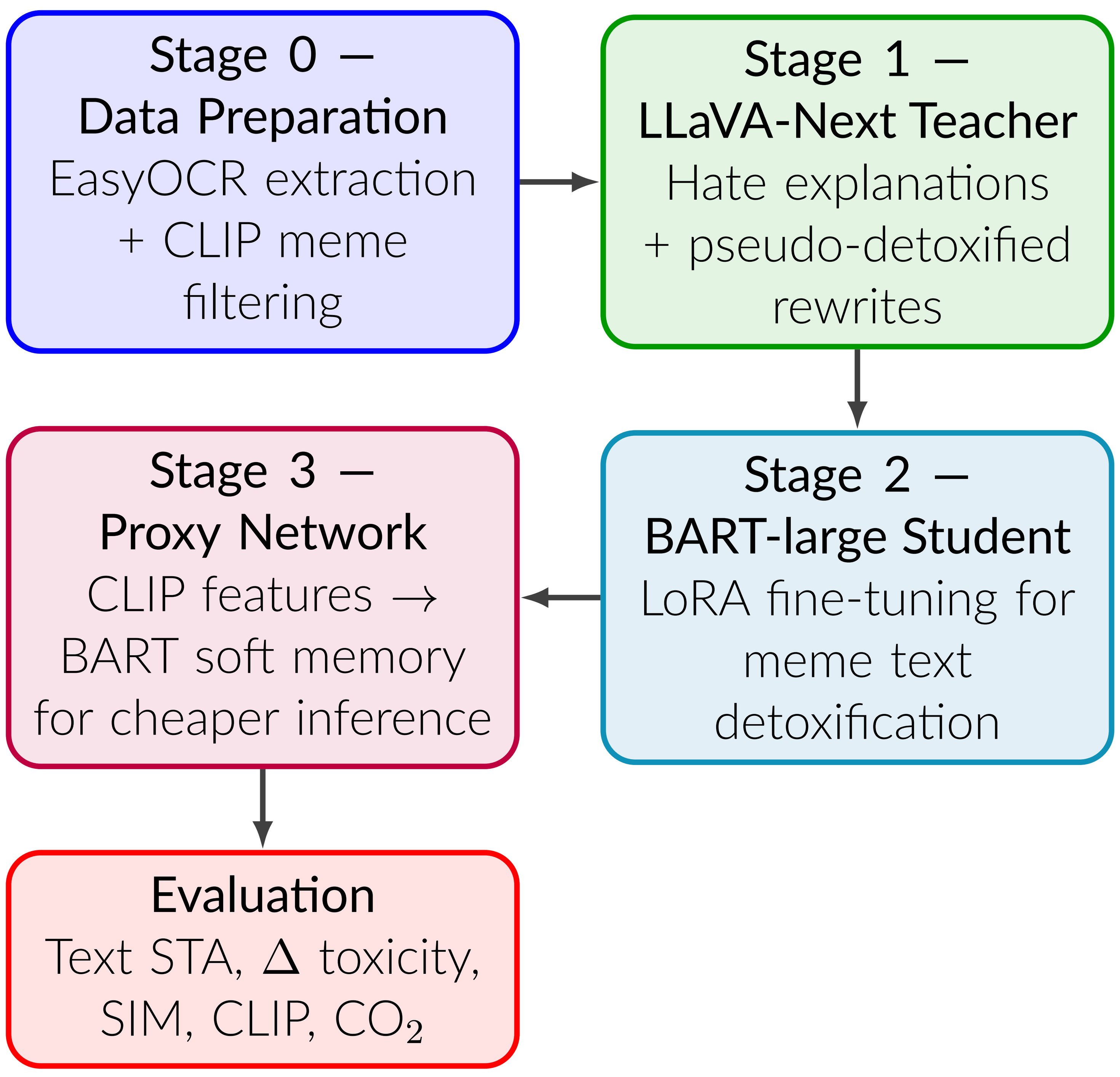
This project instead proposes **text detoxification**: rewriting harmful meme text in real time while preserving the core meaning and visual coherence of the original meme. The goal is proactive moderation without erasure, enabling safer message delivery rather than simple deletion.

The research hypothesis is that a large multimodal teacher can explain why a meme is harmful and generate safer pseudo-rewrites, then transfer this context-aware rewriting behavior to a compact BART-large student for efficient deployment.

Related works

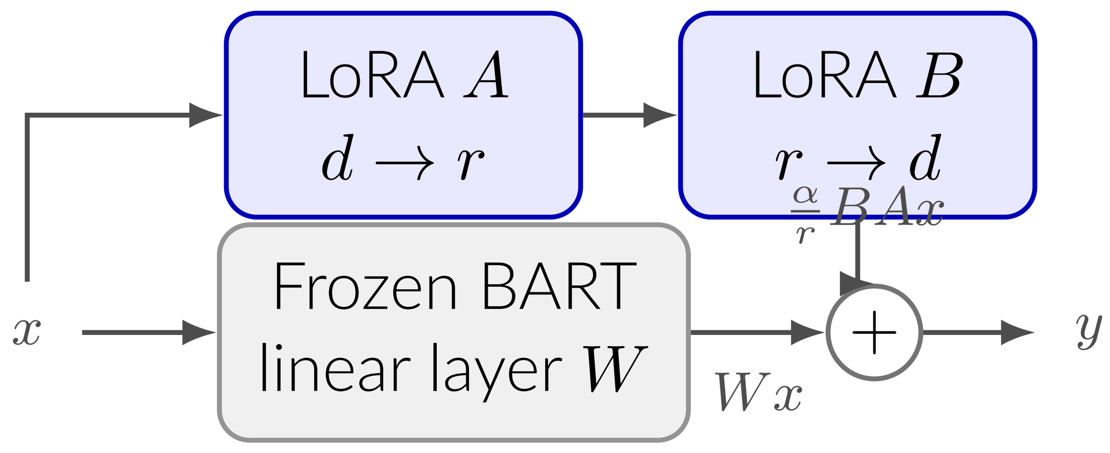
- **Hateful meme detection.** The dominant literature frames harmful memes as a classification task. The Hateful Memes challenge showed that multimodal fusion models outperform unimodal baselines, and later VisualBERT- and CLIP-based systems further improved image–text reasoning [?, ?, ?]. These methods detect or remove content, but do not preserve the original communicative intent through rewriting.
- **Text detoxification and style transfer.** ParaDetox treats toxicity as a style attribute and trains sequence-to-sequence models such as BART/T5 to neutralize toxic sentences [?]. However, plain-text detoxification ignores the image context that often gives meme text its harmful meaning.
- **Our gap.** We use LLaVA-Next's multimodal understanding as a teacher to produce target groups, visual evidence, implicit meanings, and pseudo-detoxified rewrites [?]. These labels are distilled into a lightweight BART-large student, enabling context-aware rewriting at much lower inference cost.

Method



- **Stage 0:** EasyOCR extracts meme text; CLIP filters non-meme-like images, using stricter negative prompts for MMHS150K.
- **Stage 1:** LLaVA-Next runs in two sharded passes: first structured hate explanations, then pseudo-detoxified rewrites.
- **Stage 2:** noisy pseudo-labels are filtered by parse validity, Stage 1 quality checks, positive toxicity reduction, and text quality before LoRA fine-tuning BART-large.
- **Stage 3:** a proxy MLP maps CLIP image/text features to a short BART encoder-memory sequence, exploring a VLM-free deployment path.

Parameter-efficient adaptation with LoRA. BART-large is kept frozen while small trainable low-rank matrices are added to the attention and feed-forward projections. Each adapted layer becomes $W' = W + \frac{\alpha}{r}BA$, so training updates only the adapter path rather than the full 406M-parameter model.



Only the low-rank adapters are updated; the 406M BART weights stay frozen. Here $r = 32$, $\alpha = 64$, giving 17M trainable parameters.

LoRA setup $r = 32$, $\alpha = 64$, dropout 0.05; 17M trainable parameters (~4.3%)

Datasets

HarMeme	~3,500 memes related to COVID-19 and US politics, manually labelled as harmful/non-harmful with associated textual claims [?].
MAMI	10,000 misogynist memes from Twitter [?].
MMHS150K	~150,000 heterogeneous and noisy Twitter image–text posts [?].

All datasets are filtered to retain meme-like images only. The final Stage 2 supervision contains 4,127 clean pseudo-rewrite pairs from 7,918 loaded LLaVA rows. These are split into **3,578 train**, **269 validation**, and **280 test** examples. The train split fine-tunes the BART student; validation drives early stopping and hyperparameter selection; the test split is held out and provides all reported metrics.

Validation

Quality on 280 held-out test memes. Text STA is the mean non-toxic probability from a RoBERTa toxicity classifier; Δ toxicity is the improvement over the original meme text; SIM is BERTScore F1; CLIP measures image–text alignment.

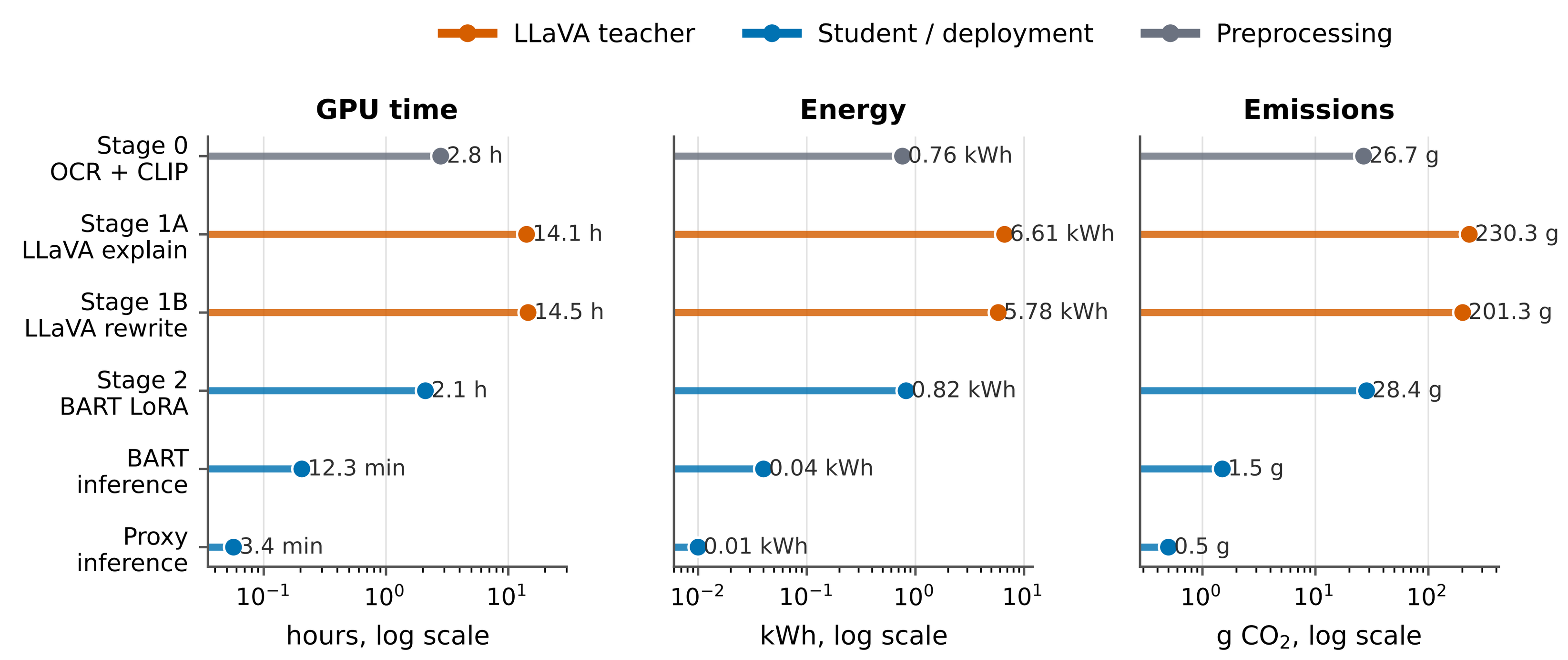
System	Text STA	Δ toxicity	SIM	CLIP
LLaVA-Next Teacher	**0.990**	**0.234**	0.472	0.636
DetoxLLM baseline	0.940	0.184	0.443	0.634
BART-large (no fine-tuning)	0.973	0.217	−0.061	0.622
BART-large FT Full (ours)	0.959	0.203	0.340	0.628
CLIP Proxy + BART (ours)	0.879	0.123	**0.600**	**0.639**

Qualitative rewrite

Rewrite
"Back in my day, men were men and women were women"

Reading the trade-off. BART-large FT Full gives the strongest student toxicity reduction and improves over DetoxLLM [?] on Text STA and Δ toxicity. Without fine-tuning, BART achieves high STA but collapses semantic similarity (SIM $= -0.06$), showing that fine-tuning is essential for meaning preservation. The CLIP Proxy reaches best-in-class SIM (0.60) at the cost of detoxification strength, revealing a clear fidelity–detoxification trade-off.

Compute and emissions.



Stage 1 dominates the one-time distillation footprint; deployed student inference is two orders of magnitude cheaper.

Stage 1 accounts for most emissions because the 7B teacher is used to create pseudo-labels. This is a one-time distillation cost: the deployed BART student is 76.6× faster than the LLaVA teacher and 16.3× faster than DetoxLLM in the repository benchmark.

+1.88 pts	76.6×	496 g
Text STA vs. DetoxLLM	BART speedup vs. LLaVA	total measured CO_2

Limitations

- Dataset noise and heterogeneity remain substantial, especially in MMHS150K.
- The student depends on LLaVA pseudo-labels; filtering improves precision but reduces the training set size.
- BART has limited direct visual grounding at inference time; image context is distilled through text fields rather than consumed as pixels.
- Evaluation is only partially multimodal. VisualBERT scores were difficult to interpret as absolute detoxification metrics in this run.
- The proxy model still shows an unbalanced toxicity–similarity trade-off.

Conclusion

This project shows that multimodal supervision from LLaVA-Next can be distilled into a lightweight BART-large student that performs meme-specific text detoxification: it reduces toxicity while preserving semantic alignment and image–text coherence.

The result supports a shift from reactive censorship to constructive rewriting. Future work should improve dataset quality, strengthen direct multimodal grounding, and integrate OCR plus inpainting modules for live rewriting of harmful text in video streams and real-world physical environments.

References